

Performance Analysis of Deepfake Text Detection Techniques on Social-media

1st Adama GaurUniversity School of Automation &
RoboticsGuru Gobind Singh Indraprastha
University, East Delhi Campus,
Delhi, India
gauradama725@gmail.com2nd Sanjay Kumar SinghUniversity School of Automation &
RoboticsGuru Gobind Singh Indraprastha
University, East Delhi Campus,
Delhi, India
sanjayksingh.012@gmail.com3rd Pranshu SaxenaDepartment of Information Technology
ABES Engineering CollegeGhaziabad, India
pranshusaxena@gmail.com

Abstract—In the last few years, there has been remarkable progress in the domain of natural language generation & understanding. This has led to the development of enhanced text generation capability of machines that can generate spurious text or deepfake text. The technology may be exploited to change and shape public sentiment on social media. Therefore, mechanisms to detect deepfake text is a crucial task. The study emphasizes on detection of deepfake text on social media platform X (formerly Twitter). It presents the performance of machine learning classifiers Logistic regression (LR), XGB classifier, and AdaBoost classifier, utilizing feature engineering methods Term Frequency (TF) and Term Frequency-Inverse Document Frequency (TF-IDF). Deep learning methods such as Dense Neural Networks (DNN), Bi-directional Long Short-Term Memory (Bi-LSTM), and Convolutional Neural Networks (CNN) in addition to state-of-the-art models and variants such as Bi-Directional Encoder Representations from Transformers (BERT), DistilBERT & RoBERTa are also a part of study, leading to 89.17% accuracy in deepfake text detection.

Keywords—Deepfake, Text Classification, Deep learning, Tweets, convolution neural networks

I. INTRODUCTION

Today social media has become an integral part of most people's lives. Developed to connect people, enabling them to share their views and thoughts via various media like text, images, audio, and video rapidly with ease, it also possesses the ability to spread fake news and provoke or mislead users by providing them with misinformation which can be disastrous for society.

The science of training a system to learn from its experiences and perform tasks without being explicitly programmed for that particular task is machine learning. Conventional machine learning requires human interference in the process of feature engineering to obtain a suitable internal representation of data for prediction [1]. Deep learning is a part of machine learning. Deep learning methods are representation-learning methods that allow a machine to automatically discover the representations required for a task from raw data [2]. A social bot is a computer algorithm that automatically generates content and interacts with humans by liking/disliking, uploading, or sharing a post on social media [3]. The word deepfake is composed of the terms “deep learning” and “fake”, referring to the synthetic multimedia generated via deep learning methods, which is often realistic. Deepfake has raised concerns, especially for people involved in politics and the film industry [4]. There have been instances of bots using ChatGPT, and GPT-3 to spread misinformation about cryptocurrency [5] and post comments on social media platform Reddit [6]. Further, a network of many such bots, known as botnets has even attempted to steal photos of real

users and converse with each other to appear more human-like [7]. People appear to be finding artificial intelligence-generated tweets more convincing than those tweeted by humans [8]. Therefore, the detection of such bots is of prime importance.

The experiment emphasizes on detection of deepfake text generated by such bots utilizing multiple machine learning & deep learning architectures including some advanced pre-trained models. The study uses the TweepFake dataset which is a publicly accessible labeled dataset having human-generated and bot-generated tweets on X [9]. The bot-generated tweets are created using different architectures like Recurrent Neural Network (RNN) and GPT-2.

The rest of the study is divided into sections: literature review in section 2, methods in section 3, result and discussion in section 4, and conclusion and future work in 5 section.

II. LITERATURE REVIEW

The term “deepfake” was coined by a Reddit user in the year 2017, who posted deepfake videos of celebrities, swapping their face with those of actresses. In the domain of computer vision, deepfake can be divided into three categories: lip sync, puppet master, and face swap [10]. Lip sync refers to the modification in the movement of the mouth under custom audio, as shown by the authors of [11]. By puppet master, facial expressions can be changed, whereas, with a face swap, the face of the target is swapped by another face.

The existence and synthesis of audio deepfake are relatively recent, enabling users to synthesize audio in the voice of people, saying things they never said [12]. In 2019, fraudsters generated the chief executive's voice of a firm and committed a fraud of \$243,000 over a call [13].

In 2013, word2vec a method to learn word embedding was introduced by researchers at Google which aimed at improving the ability of machines to represent phrases as vector representations [14]. In the following years, variants of RNN were created to address different issues such as improving memory capacity in Recurrent Neural Networks with External Memory (RNN-EM) [15] and parallelism across timesteps in Quasi-Recurrent Neural Networks (QRNN) for faster training and testing [16]. However, RNNs could not generate longer pieces of text due to the vanishing gradient problem.

In 2017, researchers introduced a neural network architecture, namely “Transformer” [17] utilizing attention mechanisms, which enhanced both natural language understanding (NLU) as well as natural language generation (NLG) in the subsequent models.

In 2019, pre-trained generative model GPT-2 architecture led to the generation of higher-quality text [18]. BERT architecture was designed to pre-train on bidirectional representations [19]. To manually detect generated text, researchers in [20] created a tool Giant Language Model Test Room (GLTR) based on statistical detection methods, raising the human detection rate of generated text to 72%.

In 2020, the authors in [21] presented a text generation model named GROVER which could generate fake news and detect it as well. The authors found that GROVER itself was best at detecting the fake news it created. In the same year, a large language model (LLM) GPT-3 was introduced by OpenAI, followed by ChatGPT in 2022 and GPT-4 in 2023 [22]. The authors in [23] attempted to identify generated text using ChatGPT and GPT-4. They found ChatGPT to be a better identifier than GPT-4 which appeared to be sensitive to noise.

In [24], researchers emphasized considering additional linguistic features and used them along with emoji embeddings to identify generated text on Twitter. Authors in [25] used Convolutional Neural Networks (CNN) with FastText embeddings which led to better accuracy of the classifier.

III. PROPOSED METHODS

This section discusses the dataset used, its pre-processing, feature engineering methodologies, machine learning, deep learning, and state-of-the-art classifiers along with the evaluation metrics used. The proposed methodology is presented in Figure 1.

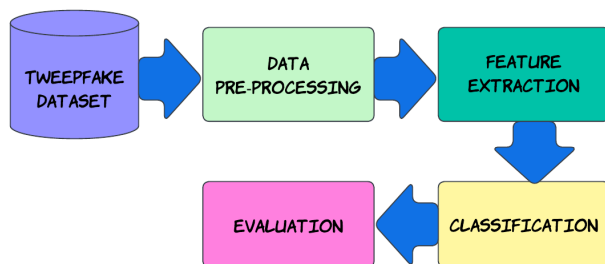


Fig. 1. Proposed Methodology

A. Dataset

For the study, the TweepFake dataset [9] has been used. The dataset consists of 25,572 tweets in total, with each tweet labeled as “human” or “bot”. Further, if a tweet is labeled as “bot”, its further class type is also mentioned. The plot in Figure 2 displays a count-plot of the account types, which indicates the absence of data imbalance.

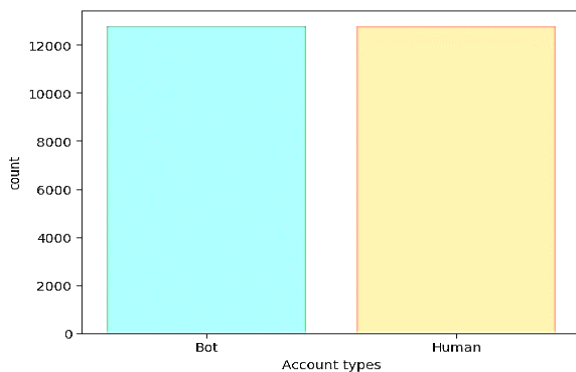


Fig. 2. Count-plot of account types

The donut displayed in Figure 3 presents the class-type distribution of tweets, indicating the architectures used in generating the deepfake text.

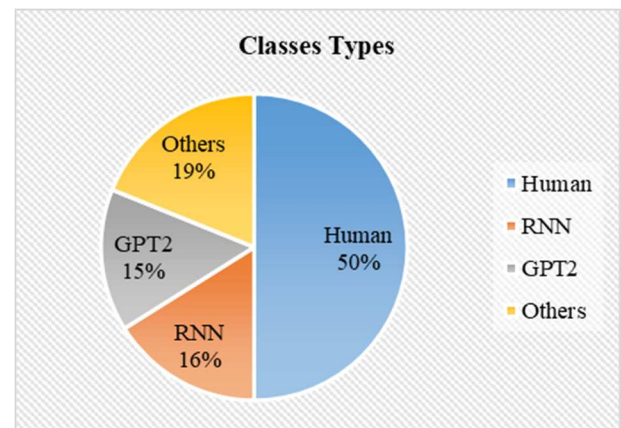


Fig. 3. Donut of class types

1) *Data Pre-processing*: To increase the performance of classifiers and save computational resources, pre-processing of data is required. The data is checked for missing values, which are none in the dataset used. To use the models, the categorical labels are converted to cardinal numbers. The tweets may be in upper case, title case, sentence case, etc. which makes a word in different cases appear different to the model. To avoid this, the tweets are lowercase.

Stopwords are the commonly used words in a language, with little or no meaning which help in forming sentences. Punctuation marks, website links, and stopwords are removed from the tweets as they do not help the model much in classifying the tweet. A word cloud of the tweets is shown in Figure 4.



Fig. 4. Wordcloud of tweets

B. Feature Extraction

Feature extraction techniques transform raw data into a new smaller set of numerical features while preserving the information present in the source data. Machine-learning models require feature extraction to produce meaningful outputs. The following feature extraction techniques are used.

1) *TF*: It indicates the count of a particular word in the given corpus. It is based on the idea that frequently appearing terms are likely to be of greater importance for the model.

2) *TF-IDF*: It is another popular feature extraction technique (eq. 1). It focuses on finding out more relevant terms in a document compared to the corpus. For a word w in a document d , it is measured using the given below equation:

$$TF - IDF(w, d) = TF(w, d) * IDF(w) \quad (1)$$

where $IDF(w)$ in eq. 2 is the inverse document frequency. Given that n is the total number of documents in the set and $DF(w)$ is document frequency, $IDF(w)$ is calculated as:

$$IDF(w) = \log\left(\frac{n+1}{DF(w)+1}\right) + 1 \quad (2)$$

C. Machine learning classifiers

This subsection discusses the machine learning classifiers used in the study. One of the most commonly used machine learning models for binary classification is Logistic Regression (LR). It involves the application of logit transformation on odds [26].

XGB or eXtreme Gradient Boosting is another popular model based on the gradient-boosted decision tree model. It is preferred due to its high computational speed with avoidance of overfitting due to regularization [27].

The Adaboost or Adaptive Boosting Classifier is a boosting technique that fits multiple copies of the classifier on the data and assigns higher weights to the instances that were misclassified to curb them.

D. Deep learning classifiers

The following deep learning classifiers are used in the study:

1) *DNN*: A DNN is one of the most simplified yet powerful deep learning architectures. A DNN consists of multiple hidden layers. It has several densely connected neurons, where each neuron of a layer accepts input from every neuron available in the former layer, whose neurons self-optimize during the learning process [28]. The DNN used in the study utilizes 4 such dense layers with ReLU and sigmoid as activation functions in the inner & outer layers respectively under the binary classification nature of the task.

2) *CNN*: CNN architectures are a type of deep neural network utilizing a convolutional layer for extracting features from the data, using a learnable filter (kernel) that slides over the input vectors [29]. The CNN used in the study comprises three such convolutional layers, each of which is followed by a pooling layer. A dropout of 10% is also used to avoid overfitting by the model.

3) *Bi-LSTM*: Unlike the LSTM model, which takes the input in the forward direction only, Bi-LSTM consists of 2 LSTM layers which enables it to take the input in both directions, backward to forward and forward to backward, hence the term “Bi-directional”. The 2 LSTM layers help the model to have access to more input and understand the context more effectively. Figure 5. Displays the architecture used in the study.

E. State-of-the-art Classifiers

This section discusses the state-of-the-art classifiers & their variants used in the study.

1) *BERT*: The BERT (bi-directional encoding representations from transformers) [30] model was made by Google, which was pre-trained on representations from text from both directions, forward to backward, and vice versa. Though Bi-LSTM also uses bi-directionality, the BERT model outperformed Bi-LSTM as indicated by its authors. One possible reason is due to their attention mechanism and architecture, transformers are more capable of handling longer texts than conventional architectures like RNN and LSTM. The cased BERT model was used in the study, i.e. words were not lowercase, as it had been in the input data of previous models discussed in the study.

2) *DistilBERT*: DistilBERT is a distilled version of BERT [31] which is about 40% smaller than the original BERT model and at the same time produces 60% faster inference while retaining 97% performance of the BERT model. It uses a knowledge distillation technique, through which knowledge from a larger model is transferred to compact models, making them computationally cheaper.

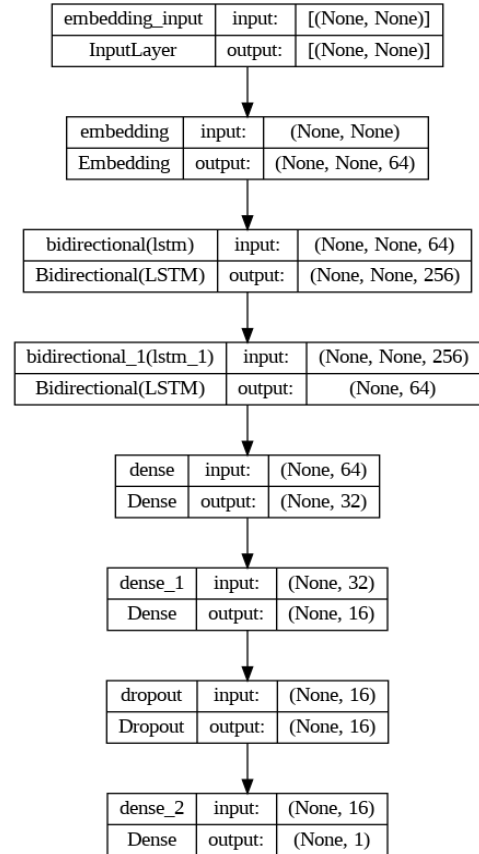


Fig. 5. Bi-LSTM architecture used

3) *RoBERTa*: A Robustly Optimized BERT Pretraining Approach, RoBERTa [32] was developed by Facebook. They made changes to the BERT model, training it on more data and longer pieces of text. The objective of the next sentence prediction was discarded and dynamic masking

during training was used. RoBERTa achieved state-of-the-art results on RACE, SQuAD, and GLUE benchmarks.

F. Evaluation Metrics

To assess and compare the performance of the model, various types of evaluation metrics are used based on the task performed. In the study, binary classification was performed by the classifiers, therefore accuracy, f1 score, precision, and recall were used as evaluation metrics.

Accuracy (eq. 3) measures how often the classifier predicts all the classes correctly. It is computed as:

$$Accuracy = \frac{TP+TN}{FP+FN+TP+TN} \quad (3)$$

Where,

TP: is known as true positive, indicating correct predictions of positive class

TN: is known as true negative, indicating correct predictions of negative class

FP: is known as false positive, indicating incorrect predictions of positive class

FN: is known as false negative, indicating incorrect predictions of negative class

Precision determines the proportion of correctly identified positive classes to the total instances of positive classes. It is calculated as (eq. 4):

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

Recall (eq. 5) helps in identifying how well the model correctly identifies the positive class. It is computed as:

$$Recall = \frac{TP}{FN+TP} \quad (5)$$

F1-score (eq. 6) is beneficial in getting the overall performance of the model, as it considers both precision and recall by taking their harmonic mean. It is computed as:

$$F1 - score = \frac{2*Precision*Recall}{Precision+Recall} \quad (6)$$

IV. RESULTS AND DISCUSSION

In the study, 9 architectures were used, which consisted of 3 machine learning models: Logistic regression, XGB classifier, and AdaBoost classifier, 3 deep learning models: DNN, CNN, Bi-LSTM, and state-of-the-art classifiers BERT, RoBERTa, and variant DistilBERT. Python 3.10.12 language was used throughout the study, using a system with Nvidia Tesla T4 GPU and Intel(R) Xeon(R) CPU @ 2.20GHz. Python libraries sci-kit-learn, TensorFlow and simple transformers were used to implement the models.

It is evident from Figure 6, that TF-IDF is better than TF, leading to greater accuracy of the models.

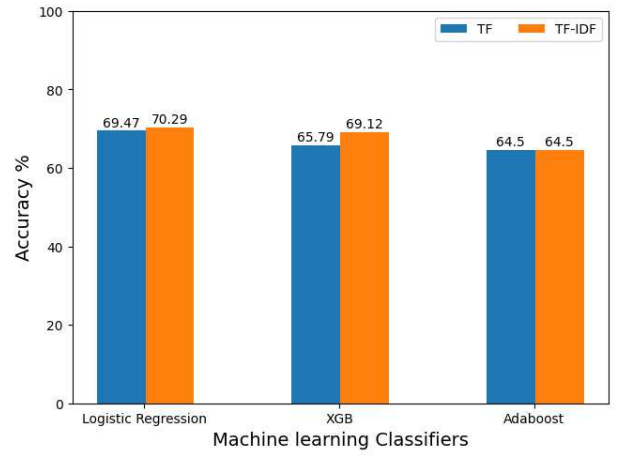


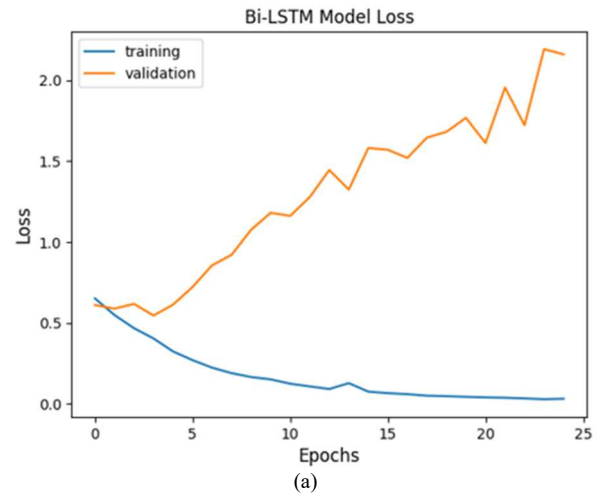
Fig. 6. Comparison of Machine learning architectures using different feature extraction techniques

Table 1 displays the evaluation metrics of the classification architectures used in the study. Machine learning architectures utilizing the TF-IDF technique are taken into consideration further. Though the BERT model achieves the highest accuracy, f1 score, and precision, it has lower recall.

TABLE I. EVALUATION RESULTS OF THE MODELS

Classifier	Accuracy	F1 Score	Precision	Recall
LR	0.703	0.717	0.684	0.753
XGB	0.691	0.731	0.647	0.842
Adaboost	0.645	0.727	0.59	0.948
DNN	0.706	0.716	0.691	0.743
CNN	0.668	0.697	0.64	0.765
Bi-LSTM	0.713	0.719	0.705	0.734
RoBERTa	0.878	0.882	0.857	0.908
BERT	0.892	0.891	0.895	0.887
DistilBERT	0.88	0.881	0.877	0.884

Amongst DNN, CNN, and Bi-LSTM, Bi-LSTM showed better overall performance. The accuracy and loss curves are displayed in Figure 7.



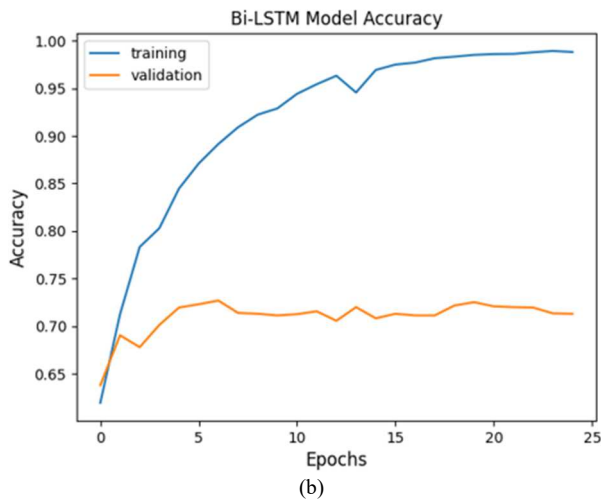


Fig. 7. Results of Bi-LSTM model. a) Loss curve b) Accuracy curve

Figure 8 shows the comparison of models based on their f1-score.



Fig. 8. Comparison of f1 scores

V. CONCLUSION AND FUTURE WORK

The study aimed at detecting deepfake text which is one of the major issues in today's era. From the study, we found that using TF-IDF with machine learning techniques proved to be a good choice, which even performed better than some deep learning models. However, the transformer architectures were best at identifying deepfake text. Considering the BERT model's performance, models using a structure similar to BERT may provide even better performance. Different feature engineering techniques and hyperparameter tuning of the models may also enhance their performance.

REFERENCES

- [1] Amazon Web Services (2023)What's the Difference Between Machine Learning and Deep Learning?.Available at <https://aws.amazon.com/compare/the-difference-between-machine-learning-and-deep-learning/>(Accessed: 9 January 2024).
- [2] LeCun, Y., Bengio, Y. and Hinton, G., 2015. Deep learning. *nature*, 521(7553), pp.436-444.
- [3] Davis, C.A., Varol, O., Ferrara, E., Flammini, A. and Menczer, F., 2016, April. Botornot: A system to evaluate social bots. In *Proceedings of the 25th international conference companion on world wide web* (pp. 273-274).
- [4] Holliday, C., 2021. Rewriting the stars: Surface tensions and gender troubles in the online media production of digital deepfakes. *Convergence*, 27(4), pp.899-918.
- [5] Diaz, M. (2023)ChatGPT-assisted bots are spreading on social media. Available at <https://www.zdnet.com/article/chatgpt-assisted-bots-are-spreading-on-social-media/> (Accessed: 9 January 2024).
- [6] Heaven, W.D. (2020) A GPT-3 bot posted comments on Reddit for a week n and no one noticed. Available at <https://www.technologyreview.com/2020/10/08/1009845/a-gpt-3-bot-posted-comments-on-reddit-for-a-week-and-no-one-noticed/> (Accessed: 9 January 2024).
- [7] Thaler, S. (2023)Scientists found over 1,000 AI bots on X stealing selfies to create fake accounts. Available at <https://nypost.com/2023/08/24/scientists-found-1140-ai-bots-on-x-creating-fake-profiles/> (Accessed: 9 January 2024).
- [8] Hameed, M. (2023) People may be more likely to believe AI-generated tweets: Study. Available at https://www.siasat.com/people-may-be-more-likely-to-believe-ai-generated-tweets-study-2628251/#google_vignette (Accessed: 9 January 2024).
- [9] Fagni, T., Falchi, F., Gambini, M., Martella, A. and Tesconi, M., 2021. TweepFake: About detecting deepfake tweets. *Plos one*, 16(5), p.e0251415.
- [10] Saravana Ram, R., Vinoth Kumar, M., Al-shami, T.M., Masud, M., Aljuaid, H. and Abouhawwash, M., 2023. Deep Fake Detection Using Computer Vision-Based Deep Neural Network with Pairwise Learning. *Intelligent Automation & Soft Computing*, 35(2).
- [11] Suwajanakorn, S., Seitz, S.M. and Kemelmacher-Shlizerman, I., 2017. Synthesizing obama: learning lip sync from audio. *ACM Transactions on Graphics (ToG)*, 36(4), pp.1-13.
- [12] Almutairi, Z. and Elgibreen, H., 2022. A review of modern audio deepfake detection methods: challenges and future directions. *Algorithms*, 15(5), p.155.
- [13] Stupp, C. (2019) Fraudsters Used AI to Mimic CEO's Voice in Unusual Cybercrime Case. Available at <https://www.wsj.com/articles/fraudsters-use-ai-to-mimic-ceos-voice-in-unusual-cybercrime-case-11567157402> (Accessed: 9 January 2024).
- [14] Mikolov, T., Sutskever, I., Chen, K., Corrado, G.S. and Dean, J., 2013. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26.
- [15] Peng, B. and Yao, K., 2015. Recurrent neural networks with external memory for language understanding. *arXiv preprint arXiv:1506.00195*.
- [16] Bradbury, J., Merity, S., Xiong, C. and Socher, R., 2016. Quasi-recurrent neural networks. *arXiv preprint arXiv:1611.01576*.
- [17] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. *Advances in neural information processing systems*, 30.
- [18] von Davier, M., 2019. Training Optimus prime, MD: Generating medical certification items by fine-tuning OpenAI's gpt2 transformer model. *arXiv preprint arXiv:1908.08594*.
- [19] Devlin, J., Chang, M.W., Lee, K. and Toutanova, K., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- [20] Gehrmann, S., Strobelt, H. and Rush, A.M., 2019. Gltr: Statistical detection and visualization of generated text. *arXiv preprint arXiv:1906.04043*.
- [21] Zellers, R., Holtzman, A., Rashkin, H., Bisk, Y., Farhadi, A., Roesner, F. and Choi, Y., 2019. Defending against neural fake news. *Advances in neural information processing systems*, 32.
- [22] Malik, E. (2023) Artificial Intelligence (AI) and ChatGPT: history and timelines. Available at <https://www.officetimeline.com/blog/artificial-intelligence-ai-and-chatgpt-history-and-timelines> (Accessed: 9 January 2024).
- [23] Bhattacharjee, A. and Liu, H., 2023. Fighting Fire with Fire: Can ChatGPT Detect AI-generated Text?. *arXiv preprint arXiv:2308.01284*.
- [24] Chong, A.T.Y., Chua, H.N., Jasser, M.B. and Wong, R.T., 2023, October. Bot or Human? Detection of DeepFake Text with Semantic, Emoji, Sentiment and Linguistic Features. In *2023 IEEE 13th International Conference on System Engineering and Technology (ICSET)* (pp. 205-210). IEEE.
- [25] Sadiq, S., Aljrees, T. and Ullah, S., 2023. Deepfake Detection on Social Media: Leveraging Deep Learning and FastText Embeddings for Identifying Machine-Generated Tweets. *IEEE Access*.

- [26] IBM (2023) What is logistic regression?. Available at <https://www.ibm.com/topics/logistic-regression>(Accessed: 9 January 2024).
- [27] Aydin, Z.E. and Ozturk, Z.K., 2021. Performance analysis of XGBoost classifier with missing data. *Manchester Journal of Artificial Intelligence and Applied Sciences (MJAIAS)*, 2(02), p.2021.
- [28] Nazari, F. and Yan, W., 2021. Convolutional versus dense neural networks: Comparing the two neural networks performance in predicting building operational energy use based on the building shape. *arXiv preprint arXiv:2108.12929*.
- [29] Pham, N.Q., Kruszewski, G. and Boleda, G., 2016, November. Convolutional neural network language models. In *Proceedings of the 2016 conference on empirical methods in natural language processing* (pp. 1153-1162).
- [30] Devlin, J., Chang, M.W., Lee, K. and Toutanova, K., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- [31] Sanh, V., Debut, L., Chaumond, J. and Wolf, T., 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper, and lighter. *arXiv preprint arXiv:1910.01108*.
- [32] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L. and Stoyanov, V., 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.